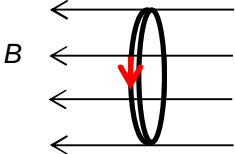


		answer
7(a)	Magnetic flux through a plane surface is defined as the product of the area and the magnetic flux density normal to the plane of the area.	[1] or zero
7(b)(i)	$\phi = BA \cos \theta$ $= 0.20 \times \pi (3.0 \times 10^{-2})^2 \cos 0^\circ$ $= 5.7 \times 10^{-4} \text{ Wb}$	[1] for correct substitution [1] for correct answer
7(b)(ii) 1.	 The diagram shows a circular loop oriented vertically. Four horizontal arrows labeled 'B' point from right to left, representing a uniform magnetic field. A red arrow on the left side of the loop points downwards, indicating the direction of the induced current.	[1] correct direction of induced current

7(b)(ii) 2.	Induced e.m.f. $E = \frac{(0.60 - 0.20)\pi(3.0 \times 10^{-2})^2}{(0.20)}$ $= 5.655 \times 10^{-3} \text{ V}$ Induced current $I = \frac{5.655 \times 10^{-3}}{0.20}$ $= 0.028 \text{ A}$	[1] value [1] ans
7(c)	When the loop is set to swing between X and X', there is a <u>change in magnetic flux</u> threading through the loop, causing an <u>induced current to flow</u> in the loop. The induced current flows in a direction to <u>oppose the change in flux</u> and hence <u>produce an opposing force/ opposing magnetic field to the motion of the loop.</u>	[1] [1]
8(a)	Solar power collected = Intensity $\times$ Area	[1] for power